

4.1 Forces: Turning Effects & Equilibrium

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.1 Forces: Turning Effects & Equilibrium
Difficulty	Medium

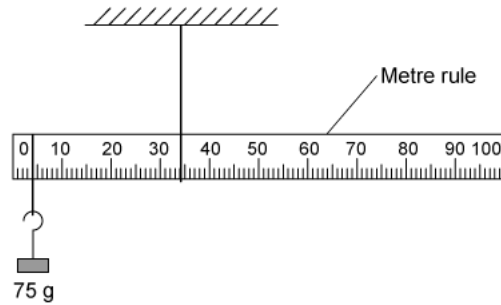
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A uniform metre rule is hung from the ceiling as shown



The rule balances when a 75 g mass is hung from one end of the ruler

What is the mass of the metre rule?

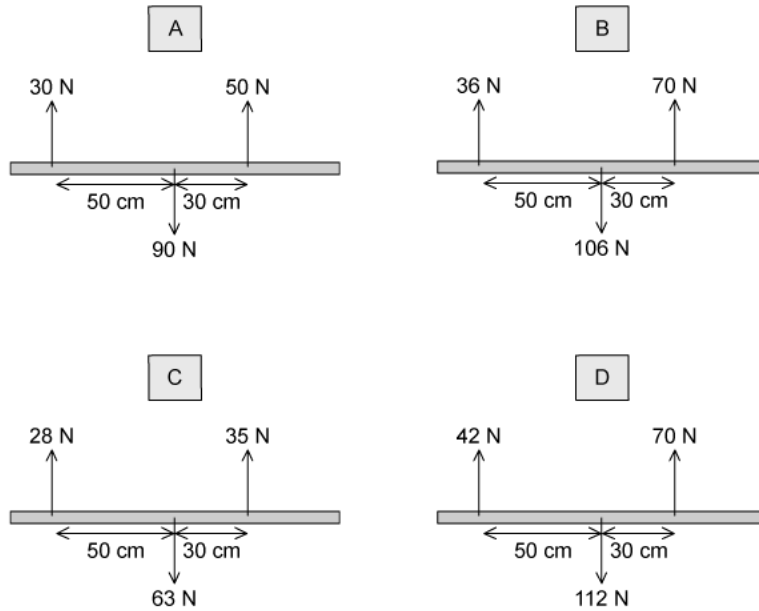
- A. 38 g
- B. 51 g
- C. 141 g
- D. 159 g

[1 mark]

Question 2

Four beams of the same length each have three forces acting on them.

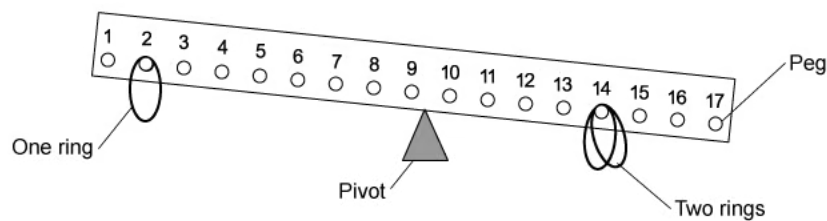
Which beam has both zero resultant force and zero resultant torque acting?



[1 mark]

Question 3

The diagram shows a child's balancing game.



The wooden rod is uniform and all the rings are of equal mass. Two rings are hung on peg 14 and one on peg 2.

On which hook must a fourth ring be hung in order to balance the rod?

- A. 2
- B. 3
- C. 5
- D. 6

[1 mark]

Question 4

An object, made from two equal masses joined by a light rod, falls with uniform speed through air. The rod remains horizontal.

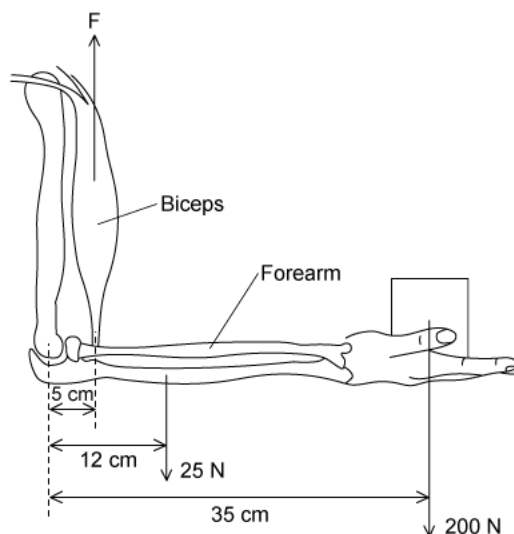
Which statement about this system is correct?

- A. it is not in equilibrium because it is falling steadily
- B. it is not in equilibrium because it is in motion
- C. it is not in equilibrium because there is a resultant torque
- D. it is in equilibrium because there is no resultant force and no resultant torque

[1 mark]

Question 5

A man holds a 200 N load stationary in his hand. The combined weight of the forearm and hand is 25 N. The forearm is held horizontal, as shown



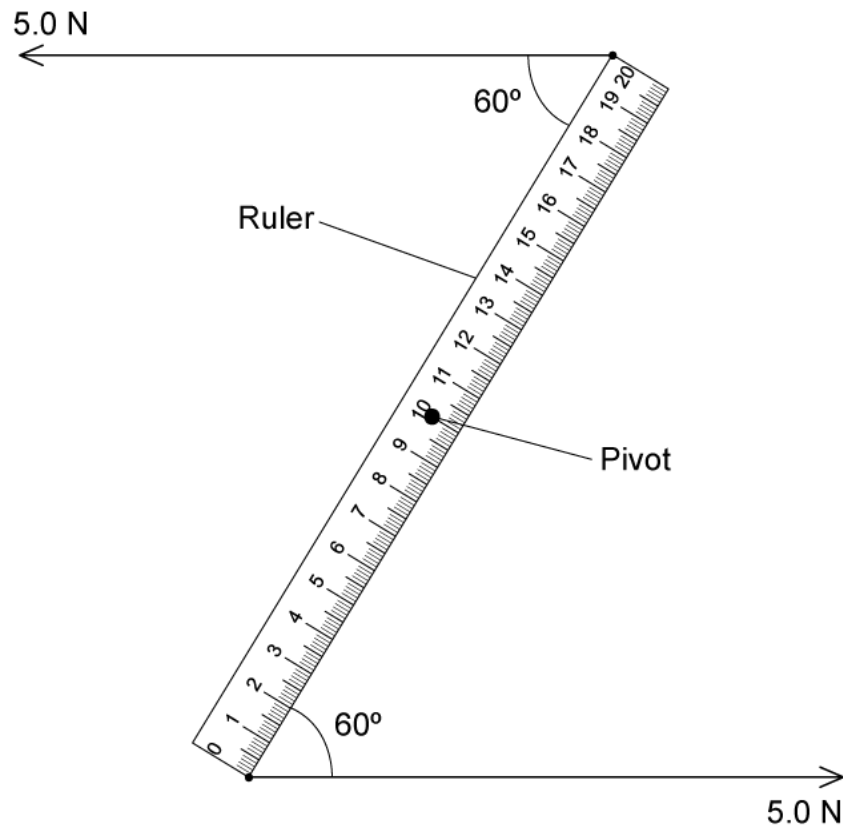
What is the vertical force F needed in the biceps?

- A. 1340 N
- B. 1400 N
- C. 1460 N
- D. 1520 N

[1 mark]

Question 6

A 20 cm ruler is pivoted at its centre. Equal and opposite forces of magnitude 5.0 N are applied to the ends of the ruler, creating a couple as shown



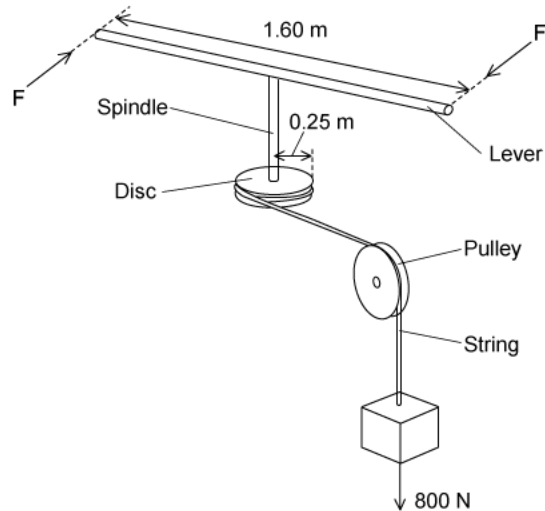
What is the magnitude of the torque of the couple on the ruler when it is in the position shown?

- A. 0.17 N m
- B. 0.87 N m
- C. 1.0 N m
- D. 1.7 N m

[1 mark]

Question 7

A spindle is attached at one end to the centre of a lever of length 1.60 m and at its other end to the centre of a disc of radius 0.25 m. A string is wrapped round the disc, passes over a pulley and is attached to an 800 N weight.



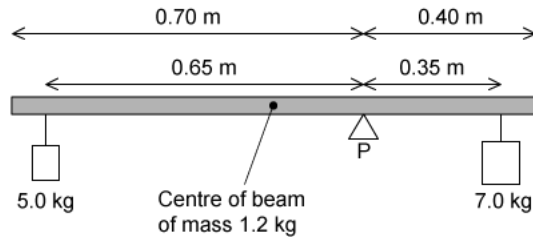
What is the minimum force F , applied to each end of the lever, that could lift the weight?

- A. 62.5 N
- B. 125 N
- C. 250 N
- D. 750 N

[1 mark]

Question 8

A uniform beam of mass 1.2 kg is pivoted at P as shown. The beam has a length of 0.70 m and P is 0.40 m from one end. Loads of 5.0 kg and 7.0 kg are suspended 0.65 m and 0.35 m from the pivot as shown.



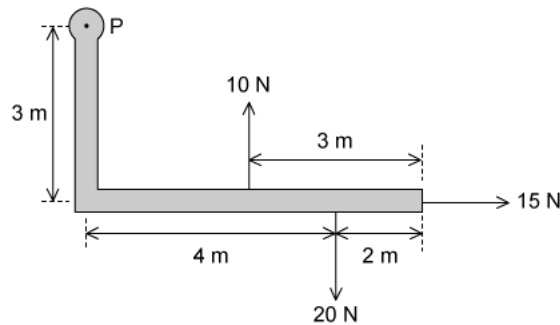
What torque must be applied to the beam in order to maintain it in equilibrium?

- A. 0.62 N m
- B. 0.98 N m
- C. 6.1 N m
- D. 9.6 N m

[1 mark]

Question 9

A rigid L-shaped lever arm is pivoted at point P.



Three forces act on the lever arm, as shown in the diagram.

What is the magnitude of the resultant moment of these forces about point P?

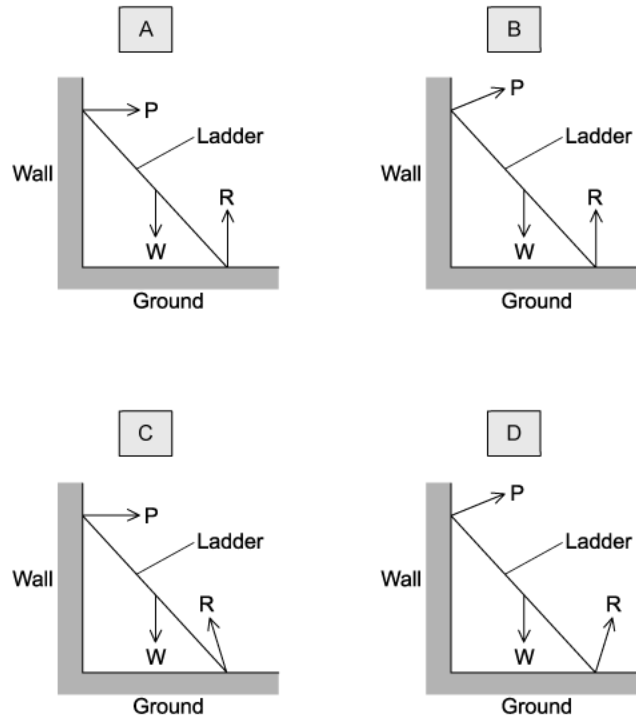
- A. 5 N m
- B. 65 N m
- C. 95 N m
- D. 155 N m

[1 mark]

Question 10

A ladder is positioned on icy ground and is leant against a rough wall. At the instant of release, it begins to slide.

Which diagram correctly shows the directions of the forces P , W and R acting on the ladder as it slides?



[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- To calculate moments, we will need to use
 - *$\text{Moment} = \text{Force} \times \text{perpendicular distance from pivot}$*
- The pivot is positioned at 34 cm
- The metre rule is uniform, meaning its weight acts at the centre mark, 50 cm
- Therefore, the distance between the pivot and the centre of gravity is
 - $50 - 34 = 16 \text{ cm}$
- The weight of the ruler produces a clockwise moment of
 - $16 \times M \times g = 16 Mg$
- The 75g mass acts downwards at the 4cm marker, which is $34 - 4 = 30 \text{ cm}$ from the pivot, producing an anticlockwise moment of
 - $75 \times g \times 30$
- For the system to be in equilibrium, clockwise moment = anticlockwise moment
 - $16 Mg = 75 \times g \times 30$
 - $M = \frac{75 \times 30}{16} = 141 \text{ g}$
- Therefore, the metre rule has a mass of 141 g

Exam Tip: when dealing with systems in equilibrium (e.g. moments, conservation of momentum) you don't need to convert units into SI units, you just need to make sure the units are the same on both sides and you need to check your final answer is reported in the correct units, take this question as an example:

The mass is reported in g so the answer must be in g

$$M = \frac{75g \times 30cm}{16cm} = 141g$$

You do not need to convert cm to m because the units cancel

2

The correct answer is **D** because:

- Torque is the moment of a force, it causes a turning effect
- Consider option **D**
- Upward force = $42 + 70 = 112 \text{ N}$
- Downward force = 112 N
 - Resultant force = $112 - 112 = 0$
- Clockwise moment = $42 \times 50 = 2100 \text{ N cm}$
- Anticlockwise moment = $70 \times 30 = 2100 \text{ N cm}$
 - Resultant moment = $2100 - 2100 = 0$
- Both the resultant force and moments are zero, so **D** is correct

$$\text{Upward force} = 30 + 50 = 80 \text{ N}$$

$$\text{Downward force} = 90 \text{ N}$$

$$\cdot \text{ Resultant force} = 90 - 80 = 10 \text{ N downward}$$

$$\text{Clockwise moment} = 30 \times 50 = 1500 \text{ N cm}$$

$$\text{Anticlockwise moment} = 50 \times 30 = 1500 \text{ N cm}$$

$$\cdot \text{ Resultant moment} = 1500 - 1500 = \mathbf{0 \text{ N cm}}$$

$$\text{Upward force} = 36 + 70 = 106 \text{ N}$$

A is incorrect because

Downward force = 106 N

· Resultant force = $106 - 106 = 0$

Clockwise moment = $36 \times 50 = 1800 \text{ N cm}$

Anticlockwise moment = $70 \times 30 = 2100 \text{ N cm}$

· Resultant moment = $2100 - 1800 = \mathbf{300 \text{ N cm}}$

Upward force = $28 + 35 = 63 \text{ N}$

Downward force = 63 N

· Resultant force = $63 - 63 = 0$

Clockwise moment = $28 \times 50 = 1400 \text{ N cm}$

Anticlockwise moment = $35 \times 30 = 1050 \text{ N cm}$

· Resultant moment = $1400 - 1050 = \mathbf{350 \text{ N cm}}$

B is incorrect because

C is incorrect because

3

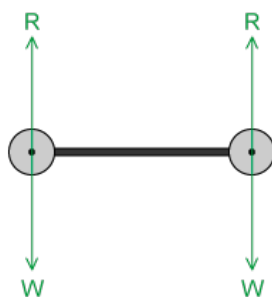
The correct answer is **D** because:

- The weight of one ring is equal to mg
- In order for the rod to be in equilibrium :
- Clockwise moments = Anticlockwise moments
 - Clockwise moment = $2 \times (14 - 9)mg = 10 \text{ mg}$
- Let x be the position of the hook where the fourth ring will be hung
 - Anticlockwise moment = $(9 - 2)mg + (9 - x)mg$
- For equilibrium:
 - $7 \text{ mg} + (9 - x)mg = 10 \text{ mg}$
 - $(9 - x)mg = 3 \text{ mg}$
 - $(9 - x) = 3$
 - $x = 9 - 3 = 6$

4

The correct answer is **D** because:

- For an object to be in equilibrium, it must be experiencing no acceleration. This means that both the net force and the net torque on the object must be zero
- Since the masses are both equal, their weight will be the same and pointing downwards, so the clockwise and anticlockwise moments will be the same, therefore their resultant torque must be zero
- We are told the object moves with uniform speed, or terminal velocity, this means the air resistance on the object will be same as the weight and hence there is no acceleration or resultant force
- Therefore, the object satisfies both conditions for equilibrium



A & B are incorrect because both these statements argue that the object cannot be in equilibrium if the object is moving, however, this is not true, as long as an object is not accelerating then it can be in equilibrium

C is incorrect because a resultant torque could only occur if the forces were unbalanced, however, the object moves with constant speed so the forces must all be balanced

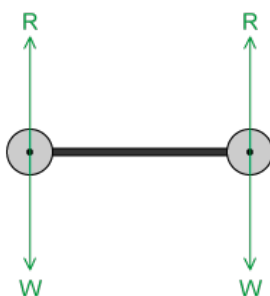
5

The correct answer is **C** because:

- We are told the mass is stationary in this person's hand, therefore, the system is in equilibrium
- For a system in equilibrium :
 - Sum of clockwise moments = Sum of anticlockwise moments
- Clockwise moments = $(25 \times 12) + (200 \times 35) = 7300 \text{ N cm}$

The correct answer is **D** because:

- For an object to be in equilibrium, it must be experiencing no acceleration. This means that both the net force and the net torque on the object must be zero
- Since the masses are both equal, their weight will be the same and pointing downwards, so the clockwise and anticlockwise moments will be the same, therefore their resultant torque must be zero
- We are told the object moves with uniform speed, or terminal velocity, this means the air resistance on the object will be same as the weight and hence there is no acceleration or resultant force
- Therefore, the object satisfies both conditions for equilibrium



A & B are incorrect because both these statements argue that the object cannot be in equilibrium if the object is moving, however, this is not true, as long as an object is not accelerating then it can be in equilibrium

C is incorrect because a resultant torque could only occur if the forces were unbalanced, however, the object moves with constant speed so the forces must all be balanced

5

The correct answer is **C** because:

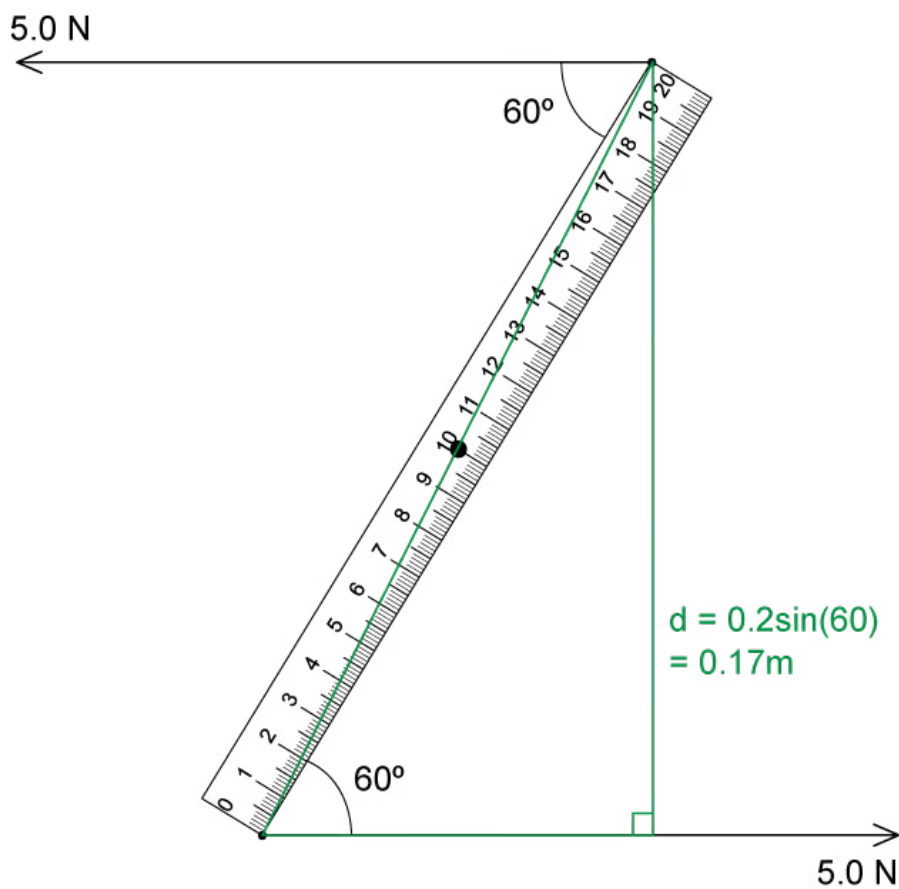
- We are told the mass is stationary in this person's hand, therefore, the system is in equilibrium
- For a system in equilibrium :
 - Sum of clockwise moments = Sum of anticlockwise moments
- Clockwise moments = $(25 \times 12) + (200 \times 35) = 7300 \text{ N cm}$
- Anticlockwise moments = $5F$
 - $5F = 7300$
 - $F = \frac{7300}{5} = 1460 \text{ N}$

5

6

The correct answer is **B** because:

- A couple refers to two parallel forces that are equal in magnitude, parallel but opposite in direction and are at a **perpendicular** distance from each other
- The torque of a couple is found by calculating
 - *Torque = Force $\times$ Perpendicular distance between forces*
- Perpendicular distance between forces = $0.2 \sin 60 = 0.17 \text{ m}$
- Magnitude of couple = $5.0 \times (0.2 \sin 60) = 0.87 \text{ N m}$



7

The correct answer is **B** because:

- Distance of force F from the centre of the spindle = $\frac{1.60}{2} = 0.80 \text{ m}$
- The forces are shown to be applied in a clockwise motion

- Clockwise moment $= (F \times 0.80) + (F \times 0.80) = 1.60 F$
- The 800 N weight acts at a distance 0.25 m from the spindle which causes an anticlockwise moment
 - Anticlockwise moment $= 800 \times 0.25$
- The minimum force that can be applied when the clockwise moments are equal to the anticlockwise moments
 - $(F \times 0.80) + (F \times 0.80) = 800 \times 0.25$
 - $1.60 F = 200$
 - $F = \frac{200}{1.6} = 125 \text{ N}$

8

The correct answer is **D** because:

- To calculate moments, we will need to use
 - *Moment = Force $\times$ perpendicular distance from pivot*
- Let the torque be equal to T , and assume it acts in a clockwise direction unless it is calculated to have a negative sign (which would tell us it acts in an anticlockwise direction)
- Clockwise moments:
 - $(0.35 \times 5.0 \times g) + T$
- The metre rule is uniform, meaning its weight acts at the centre mark. The length of the beam is 1.10 m, so the centre mark is at 0.55 m which is 0.15 m from the pivot
- Anticlockwise moments:
 - $(0.15 \times 1.2 \times g) + (0.65 \times 2.5 \times g)$
- For the system to be in equilibrium, clockwise moment = anticlockwise moment
 - $(0.15 \times 1.2 \times g) + (0.65 \times 5.0 \times g) = (0.35 \times 7.0 \times g) + T$
 - $0.18g + 3.25g = 2.45g + T$
 - $T = 0.98g = 9.6 \text{ N m}$
- Therefore, the torque required to keep the beam in equilibrium is 9.6 N m clockwise

9

The correct answer is **A** because:

- The force which will cause the lever to rotate clockwise is the 20 N force
- The forces which will cause the lever to rotate anticlockwise are the 10 N and 15 N forces

- Perpendicular distance of 10 N force : $(4 + 2) - 3 = 3 \text{ m}$
- Perpendicular distance of 15 N force : 3 m
- Perpendicular distance of 20 N force : 4 m
- Magnitude of resultant moment = Clockwise moment – Anticlockwise moment
- Magnitude of resultant moment = $(20 \times 4) - (10 \times 3) - (15 \times 3) = 5 \text{ N m}$

10

The correct answer is **B** because:

- The weight W always acts downwards
- The reaction force R acts in the opposite direction to the weight, upwards
- The ground is said to be icy which means it is **frictionless**, so there are no additional forces acting or opposing the motion
- Before the ladder begins to slide, there is a horizontal force P acting to the right where the ladder comes into contact with the rough wall because the ladder exerts a force to the left on the wall
- As the ladder begins to slide, the point of contact of the ladder with the wall begins to move down the wall
- We are told the wall is rough, meaning friction will oppose the motion of the ladder as it begins to move
- So, at the point of contact, P is now acting to the right along with the upwards frictional force, which leads to a resultant force as shown in diagram B

A & C are incorrect because these diagrams show the force P as horizontal on the ladder as before it begins to slide there is initially no friction force acting, however the question asks about the forces on the ladder **as** it begins to slide

C & D are incorrect because these diagrams show that there is a frictional force acting on the ground causing the resultant force of R to change direction, but this is incorrect since the ground is icy, meaning it is frictionless

